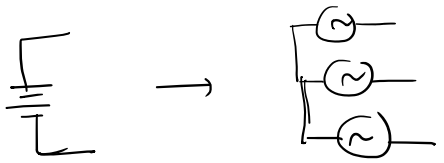
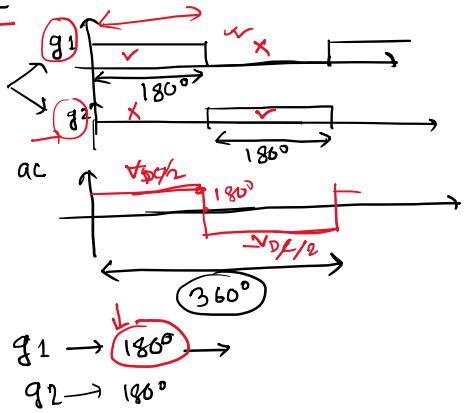
# 3-phase Inverter

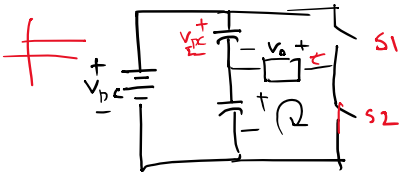
Wednesday, April 29, 2026 9:43 PM



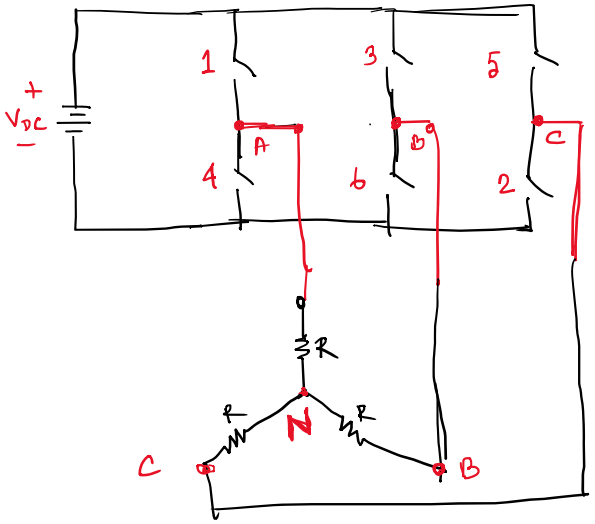
# Gate sequence




Half - bridge

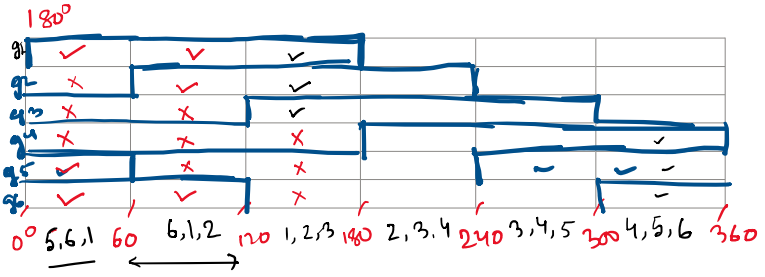



##

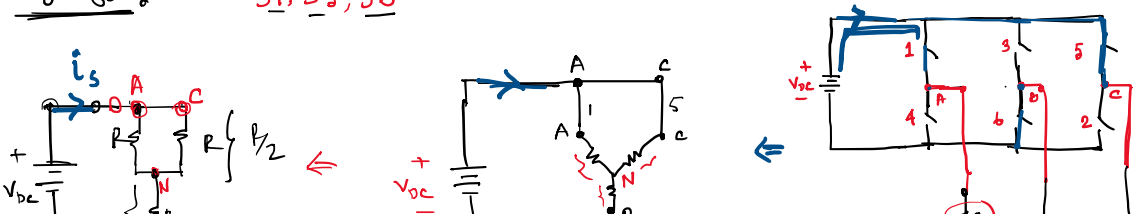


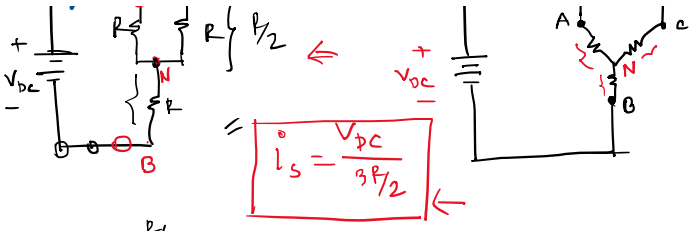

A.B,C → 120°

180° → 180°
120°




0 — 60° :- S1, S5, S6





$$i_s = \frac{V_{DC}}{3R/2}$$

$$V_{an} = \frac{R/2}{R+R/2} \times V_{DC} = \frac{V_{DC}}{3} = V_{cn}$$

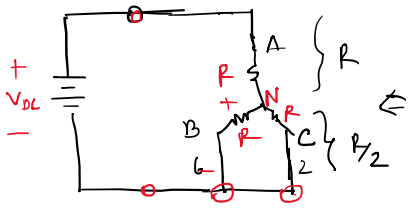
$$V_{nb} = \frac{R}{R+R/2} \times V_{DC} = \frac{2V_{DC}}{3}$$

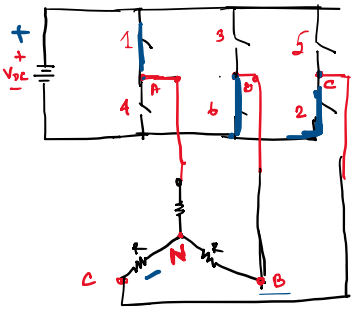
$$V_{bn} = -\frac{2V_{DC}}{3}$$

$$\begin{cases} V_{AB} = V_{DC} \\ V_{BC} = -V_{DC} \\ V_{CA} = 0 \end{cases}$$

$V_{AB} = V_{AN} - V_{BN}$

**60° – 120°:** S6, S1, S2





$$V_{an} = \frac{R}{R+R/2} \times V_{DC} = \frac{2V_{DC}}{3}$$

$$V_{nb} = \frac{R/2}{R+R/2} \times V_{DC} = \frac{V_{DC}}{3}$$

$$V_{bn} = -\frac{V_{DC}}{3} = V_{cn}$$

$V_{AB} = V_{DC}$

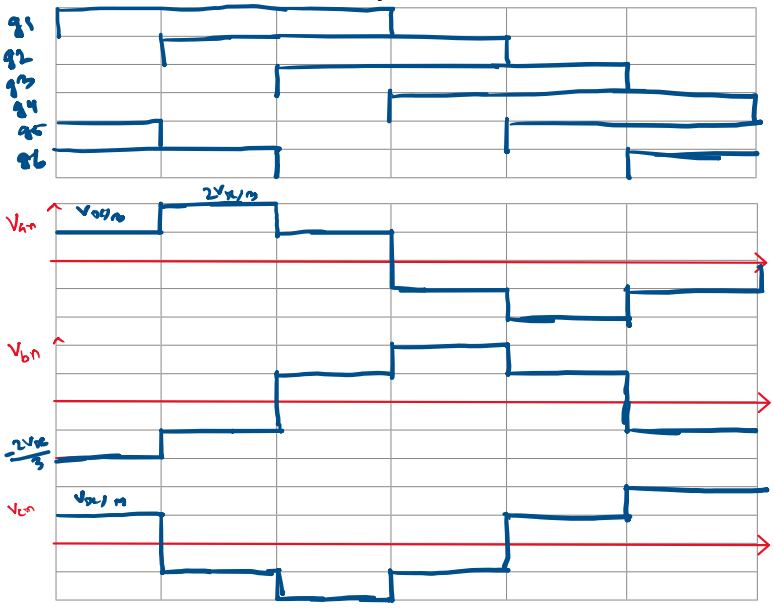
$V_{BC} = 0$

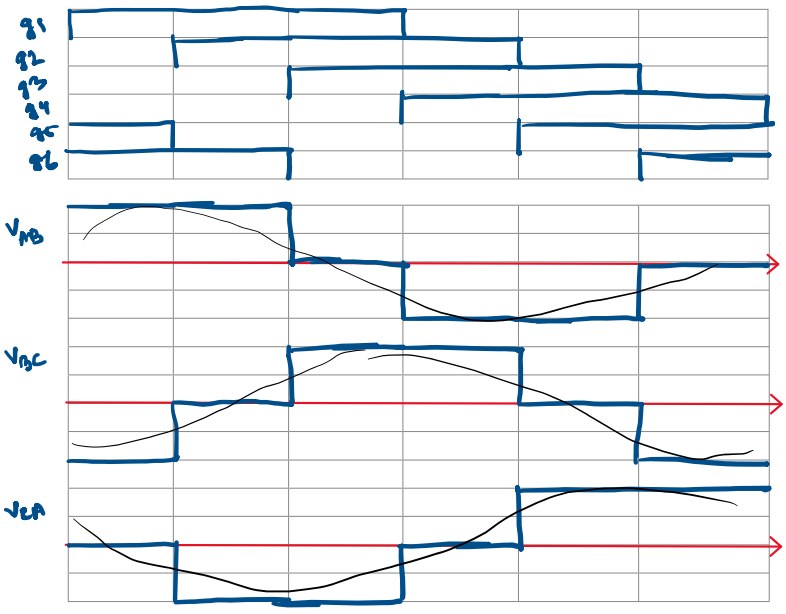
$V_{CA} = -V_{DC}$

**120° – 180°** $V_{an} = \frac{V_{DC}}{3} = V_{bn}$

$V_{cn} = -\frac{2V_{DC}}{3}$

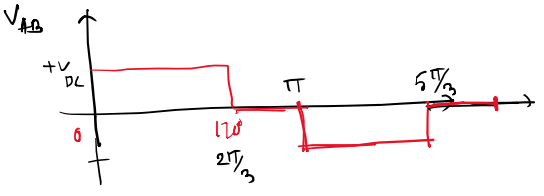
$V_{AB} = 0$ $V_{BC} = V_{DC}$ $V_{CA} = -V_{DC}$




##

$$V_{L,rms}^2 = \frac{1}{2\pi}\left[\int_0^{\frac{2\pi}{3}} V_{DC}^2\, d\theta + \int_{\pi}^{\frac{5\pi}{3}} (-V_{DC})^2\, d\theta\right]$$



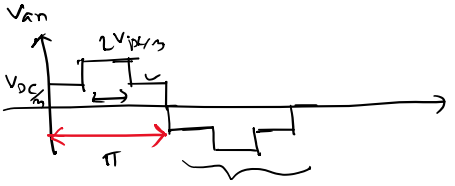

$$= \frac{1}{2\pi} \cdot \left[V_{DC}^2 \cdot \frac{2\pi}{3} + V_{DC}^2 \cdot \frac{2\pi}{3}\right]$$

$$= \frac{1}{2\pi} \cdot V_{DC}^2 \cdot \frac{2\pi}{3} \times 2$$

$$V_{L,rms}^2 = \frac{2}{3} V_{DC}^2$$

$$V_{L,rms} = \sqrt{\frac{2}{3}} V_{DC} = 0.8165\, V_{DC}$$

## $V_{an}$




$$V_{an,rms}^2 = \frac{1}{\pi}\left[\int_0^{\pi/3}\left(\frac{V_{DC}}{3}\right)^2 d\theta + \int_{\pi/3}^{2\pi/3}\left(\frac{2V_{DC}}{3}\right)^2 d\theta + \int_{2\pi/3}^{\pi}\left(\frac{V_{DC}}{3}\right)^2 d\theta\right]$$

$$= \frac{1}{\pi}\left[\frac{V_{DC}^2}{9}\cdot\frac{\pi}{3} + \frac{4V_{DC}^2}{9}\cdot\frac{\pi}{3} + \frac{V_{DC}^2}{9}\cdot\frac{\pi}{3}\right]$$

$$= \frac{V_{DC}^2}{27}\cdot(1+4+1)$$

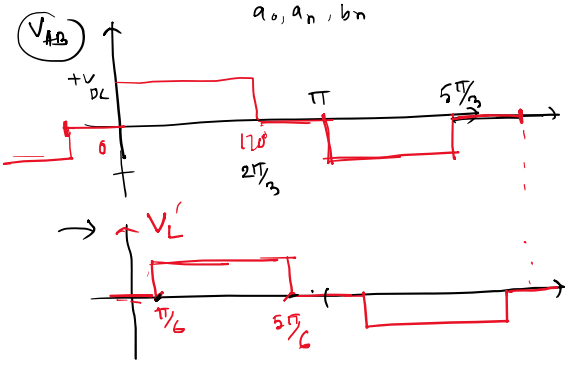
$$= \frac{6}{27}V_{DC}^2 = \frac{2}{9}V_{DC}^2$$

$$V_{an,rms} = \frac{\sqrt{2}}{3} V_{DC}$$

$$V_p = \frac{V_L}{\sqrt{3}} = \left(\sqrt{\frac{2}{3}} V_{DC}\right)\cdot\frac{1}{\sqrt{3}}$$

$$\boxed{V_p = \frac{\sqrt{2}}{3} V_{DC}}$$

## $V_{AB}$




$$V_L' = a_0 + \sum_{n=1,2}^{\infty} a_n\cos(n\omega t) + b_n\sin(n\omega t)$$

$a_0, a_n \to 0$

$$b_n = \frac{1}{T/4}\int_0^{T/2} x(t)\sin(n\omega t)\, d(\omega t)$$

$$= \frac{4}{2\pi}\int_{\pi/6}^{5\pi/6} V_{DC}\cdot\sin(n\theta)\, d\theta$$

$$= \frac{2V_{DC}}{\pi}\int_{\pi/6}^{5\pi/6}\sin(n\theta)\, d\theta$$

$$= \frac{2V_{DC}}{\pi}\left[-\frac{\cos n\theta}{n}\right]_{\pi/6}^{5\pi/6}$$

$$= \frac{2V_{DC}}{n\pi}\cdot\left[\cos n\frac{\pi}{6} - \cos n\frac{5\pi}{6}\right]$$

$$= \frac{2V_{DC}}{n\pi}\left[\cos\frac{n\pi}{6} - \cos\left\{n\left(\pi - \frac{\pi}{6}\right)\right\}\right]$$

$$= \frac{2V_{DC}}{n\pi}\left[\cos\frac{n\pi}{6} - (-1)^n\cos\frac{n\pi}{6}\right]$$

$$V_{AB} = V_L' \angle 30$$

$$\cos(A-B) = \cos A\cos B + \sin A\sin B$$

$$= \cos n\pi\cos\frac{n\pi}{6} + \sin n\pi\sin\frac{n\pi}{6}$$

($\sin n\pi = 0$)

$$= (-1)^n\cos\frac{n\pi}{6}$$

$= \frac{1}{n\pi}\left[\cos\frac{n\pi}{6} - \cdots\right]$ $= (-1)^n \cos\frac{n\pi}{6}$

$= \frac{2V_{DC}}{n\pi}\left[\cos\frac{n\pi}{6} - (-1)^n\cos\frac{n\pi}{6}\right]$

$= \begin{cases} 0 & n = \text{even} \\ \frac{4V_{DC}}{n\pi}\cos\frac{n\pi}{6} & n = \text{odd} \end{cases}$

$$V_L' = \sum_{n=1,3,5}^{\infty}\left[\frac{4V_{DC}}{n\pi}\cos\frac{n\pi}{6}\right]\sin(n\omega t)$$

$$V_{AB} = \sum_{n=1,3,5}^{\infty}\frac{4V_{DC}}{n\pi}\cos\frac{n\pi}{6}\sin\left(n\left(\omega t + \frac{\pi}{6}\right)\right)$$

$V_{BC} = V_{AB}\angle -\frac{2\pi}{3}$

$V_{CA} = V_{AB}\angle -\frac{4\pi}{3}$

$$V_{AB,n} = \frac{4V_{DC}}{n\pi}\cos\frac{n\pi}{6}$$

$$V_{AB,n\,(rms)} = \frac{4V_{DC}}{\sqrt{2}\,n\pi}\cos\frac{n\pi}{6}$$

$$V_{AB,1\,(rms)} = \frac{4V_{DC}}{\sqrt{2}\,\pi}\cos\left(\frac{\pi}{6}\right) = \frac{4V_{DC}}{\sqrt{2}\pi}\cdot\frac{\sqrt{3}}{2}$$

$$= \frac{\sqrt{6}}{\pi}V_{DC}$$

$$= 0.779\,V_{DC}$$

$$V_{AB\,(rms)} = 0.8165\,V_{DC}$$

$$THD = \frac{1}{V_{01}}\sqrt{V_0^2 - V_{01}^2}$$

\# $V_{LL} = V_{AB} = \sum_{n=1,3,5}^{\infty}\frac{4V_{DC}}{n\pi}\cos\frac{n\pi}{6}\sin\left(n\left(\omega t + \frac{\pi}{6}\right)\right)$

$$V_p = \frac{V_L}{\sqrt{3}}\angle\frac{\pi}{6}$$

positive sequence $-\pi/6$

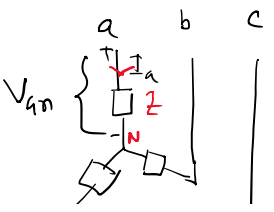
negative sequence $+\pi/6$

$$V_{an} = \sum_{n=1,3,5}^{\infty}\frac{4V_{DC}}{\sqrt{3}\,n\pi}\cos\frac{n\pi}{6}\sin\left(n\left(\omega t + \frac{\pi}{6}\right) + \frac{\pi}{6}\right)$$

\#\# Load → RL

$$I_a = \frac{V_{an}}{Z} \qquad Z = R + jn\omega L$$

$= \infty$



$= \omega$

